

Supplementary Materials

In Appendix A, we provide mathematical background. In Appendix B, we present the proofs and derivations of all the theoretical results in our paper, while Appendix C contains additional details regarding our experiments.

A. Background

Let $\nabla = (\partial/\partial x_1, \dots, \partial/\partial x_d)^\top$, $f : \mathcal{X} \rightarrow \mathbb{R}^d$ and $g : \mathcal{X} \rightarrow \mathbb{R}^{d \times p}$; the divergence operator is define as follows:

$$(\nabla \cdot f)(x) = \sum_{i=1}^d \frac{\partial f_i}{\partial x_i}(x), \quad (\nabla \cdot g)_j(x) = \sum_{i=1}^d \frac{\partial g_{ij}}{\partial x_i}(x), \quad \forall j \in \{1, \dots, p\}.$$

Expanding the term $\nabla \cdot mm^\top \nabla \log p_\theta(x)$ appearing as part of $d_m(\theta, x)$, we get

$$\begin{aligned} \nabla \cdot mm^\top \nabla \log p_\theta(x) &= \sum_{i=1}^d \frac{\partial}{\partial x_i} (mm^\top \nabla \log p_\theta(x))_i \\ &= \sum_{i=1}^d \sum_{j=1}^d \frac{\partial}{\partial x_i} ((mm^\top)_{ij} (\nabla \log p_\theta(x))_j) \\ &= \sum_{i=1}^d \sum_{j=1}^d \left(\frac{\partial}{\partial x_i} (mm^\top)_{ij} \right) (\nabla \log p_\theta(x))_j + \sum_{i=1}^d \sum_{j=1}^d (mm^\top)_{ij} (\nabla^2 \log p_\theta(x))_{ij} \\ &= \sum_{i=1}^d \sum_{j=1}^d \left(\frac{\partial}{\partial x_i} (mm^\top)_{ij} \right) (\nabla \log p_\theta(x))_j + \sum_{j=1}^d (mm^\top \nabla^2 \log p_\theta(x))_{jj} \\ &= \sum_{j=1}^d \sum_{i=1}^d \left(\frac{\partial}{\partial x_i} (mm^\top)_{ij} \right) (\nabla \log p_\theta(x))_j + \text{Tr} (mm^\top \nabla^2 \log p_\theta(x)). \end{aligned}$$

Where ∇^2 is the Hessian. The expression in the last line is more straightforward to implement in practice and it is therefore the one we use in our code.

The term $\nu(x)$ in Proposition 3.1 contains $(\nabla \cdot (mm^\top \nabla r)(x))$. The j -th index of this p -dimensional vector equals

$$(\nabla \cdot (mm^\top \nabla r)(x))_j = \sum_{i=1}^d \frac{\partial}{\partial x_i} (mm^\top \nabla r(x))_{ij}.$$

B. Theoretical Results

In this section we present the derivation of the $\mathcal{D}_m$ -posterior, along with the proof of its robustness.

B.1. Proof of Proposition 3.1

In this Subsection we present the proof of the main result of Section 3.2: the conjugacy for exponential family models.

Proof. Let p_θ be an exponential family model. Then $\nabla \log p_\theta = \nabla r(x)^\top \eta(\theta) + \nabla b(x)$, and the DSM estimator has the following form:

$$\widehat{\mathcal{D}}_m(\theta) = \frac{1}{T} \sum_{t=1}^T \underbrace{\|m^\top (\nabla r(x_t) \eta(\theta) + \nabla b(x_t))\|_2^2}_{(1)} + 2 \underbrace{\nabla \cdot (mm^\top (\nabla r(x_t) \eta(\theta) + \nabla b(x_t)))}_{(2)}.$$

Let $\stackrel{+C}{=}$ indicate equality up to an additive term that does not depend on θ .

$$\begin{aligned} (1) &= \eta(\theta)^\top \nabla r(x_t)^\top mm^\top \nabla r(x_t) \eta(\theta) + \nabla b(x_t)^\top mm^\top \nabla b(x_t) + 2\eta(\theta)^\top \nabla r(x_t)^\top mm^\top \nabla b(x_t) \\ &\stackrel{+C}{=} \eta(\theta)^\top \nabla r(x_t)^\top mm^\top \nabla r(x_t) \eta(\theta) + 2\eta(\theta)^\top \nabla r(x_t)^\top mm^\top \nabla b(x_t), \end{aligned}$$

and

$$\begin{aligned} (2) &= \nabla \cdot (mm^\top \nabla r(x_t) \eta(\theta)) + \nabla \cdot (mm^\top \nabla b(x_t)) \\ &\stackrel{+C}{=} \nabla \cdot (mm^\top \nabla r(x_t) \eta(\theta)) \\ &= \eta(\theta)^\top (\nabla \cdot (mm^\top \nabla r(x_t))). \end{aligned}$$

Therefore, $\widehat{\mathcal{D}}_m(\theta) = \eta(\theta)^\top \Lambda_T \eta(\theta) + \eta(\theta)^\top \nu_T$ where

$$\begin{aligned}\Lambda_T &:= \frac{1}{T} \sum_{t=1}^T \nabla r(x_t)^\top m m^\top \nabla r(x_t), \\ \nu_T &:= \frac{2}{T} \sum_{t=1}^T \nabla r(x_t)^\top m m^\top \nabla b(x_t) + \nabla \cdot (m m^\top \nabla r(x_t)).\end{aligned}$$

Now, assuming the prior has a p.d.f. π , the DSM-Bayes generalised posterior has a p.d.f.

$$\pi_\omega^{\mathcal{D}_m} \propto \pi(\theta) \exp(-\omega T [\eta(\theta)^\top \Lambda_T \eta(\theta) + \eta(\theta)^\top \nu_T]).$$

For $\eta(\theta) = \theta$, and the prior $\pi(\theta) \propto \exp(-\frac{1}{2}(\theta - \mu)^\top \Sigma^{-1}(\theta - \mu))$, we obtain the generalised posterior by completing the square as follows:

$$\begin{aligned}\pi_\omega^{\mathcal{D}_m}(\theta) &\propto \exp(-\frac{1}{2}(\theta - \mu)^\top \Sigma^{-1}(\theta - \mu_T)) \exp(-\omega T [\theta^\top \Lambda_T \theta + \theta^\top \nu_T]) \\ &= \exp(-\frac{1}{2}(\theta^\top \Sigma^{-1} \theta - 2\theta^\top \Sigma^{-1} \mu + \mu^\top \Sigma^{-1} \mu + \theta^\top 2\omega T \Lambda_T \theta + \theta^\top 2\omega n \nu_T)) \\ &\propto \exp(-\frac{1}{2}(\theta^\top (\Sigma^{-1} + 2\omega T \Lambda_T) \theta - 2\theta^\top (\Sigma^{-1} \mu - \omega T \nu_T))) \\ &\propto \exp(-\frac{1}{2}(\theta - \mu_T)^\top \Sigma_T^{-1}(\theta - \mu_T)),\end{aligned}$$

where

$$\begin{aligned}\Sigma_T^{-1} &:= \Sigma^{-1} + 2\omega T \Lambda_T, \\ \mu_T &:= \Sigma_T (\Sigma^{-1} \mu - \omega T \nu_T).\end{aligned}$$

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B.2. Update Parameters for online DSM-Bayes

In this Section we derive the efficient parameter updates presented in Section 3.4. To do so, we expand the expressions Λ and ν as follows:

$$\begin{aligned}\Lambda_{T+1} &:= \frac{1}{T+1} \sum_{t=1}^{T+1} \nabla r(x_t)^\top m m^\top \nabla r(x_t) \\ &= \frac{1}{T+1} \left(\sum_{t=1}^T \nabla r(x_t)^\top m m^\top \nabla r(x_t) + \nabla r(x_{T+1})^\top m m^\top \nabla r(x_{T+1}) \right) \\ &= \frac{1}{T+1} (n \Lambda_T + \nabla r(x_{T+1})^\top m m^\top \nabla r(x_{T+1})) \\ \nu_{T+1} &:= \frac{2}{T+1} \sum_{t=1}^{T+1} \nabla r(x_t)^\top m m^\top \nabla b(x_t) + \nabla \cdot (m m^\top \nabla r(x_t)) \\ &= \frac{1}{T+1} (T \nu_T + 2(\nabla r(x_{T+1})^\top m m^\top \nabla b(x_{T+1}) + \nabla \cdot (m m^\top \nabla r(x_{T+1}))))\end{aligned}$$

Now, assuming the prior has the conjugate form of Proposition 3.1, the $\mathcal{D}_m$ -posterior is given by

$$\pi_\omega^{\mathcal{D}_M}(\theta) \propto \exp(-\frac{1}{2}(\theta - \mu_T)^\top \Sigma_T^{-1}(\theta - \mu_T)),$$

where

$$\begin{aligned}\Sigma_T^{-1} &:= \Sigma^{-1} + 2\omega n \Lambda_T, \\ \mu_T &:= \Sigma_T (\Sigma^{-1} \mu - \omega n \nu_T).\end{aligned}$$

So the parameter updates for Σ_T and μ_T as T increases are:

$$\begin{aligned}\Sigma_{T+1}^{-1} &:= \Sigma^{-1} + 2\omega(T+1)\Lambda_{T+1} \\ &= \Sigma^{-1} + 2\omega(n\Lambda_T + \nabla r(x_{T+1})^\top m m^\top \nabla r(x_{T+1})) \\ &= \Sigma_T^{-1} + 2\omega \nabla r(x_{T+1})^\top m m^\top \nabla r(x_{T+1}) \\ \mu_{T+1} &:= \Sigma_{T+1} (\Sigma^{-1} \mu - \omega(T+1)\nu_{T+1}) \\ &= \Sigma_{T+1} (\Sigma^{-1} \mu - \omega(T\nu_T + 2(\nabla r(x_{T+1})^\top m m^\top \nabla b(x_{T+1}) + \nabla \cdot (m m^\top \nabla r(x_{T+1})))) \\ &= \Sigma_{T+1} (\Sigma_T^{-1} \mu_T - 2\omega(\nabla r(x_{T+1})^\top m m^\top \nabla b(x_{T+1}) + \nabla \cdot (m m^\top \nabla r(x_{T+1}))))\end{aligned}$$

obtaining the desired expression.

B.3. Global Bias-Robustness

In this Subsection, we present the theory necessary to prove that the generalized posterior presented in Section 3.1 is global bias-robust conditioned to the choice of m .

We first need to review a result from Matsubara et al. (2022a) which states conditions for a discrepancy measure $\mathcal{D}(\theta)$ in order to prove that the corresponding generalised Bayes posterior $\pi_\omega^\mathcal{D}$ is globally robust to outliers. As in the original results, we will state our findings in terms of distributions $\mathbb{P} \in \mathcal{P}(\mathcal{X})$, where $\mathcal{P}(\mathcal{X})$ denotes the set of probability distributions on $\mathcal{X}$. To this end, we will build on the notation introduced in Section 3.3, and write

$$\pi_\omega^\mathcal{D}(\theta|\mathbb{P}) \propto \pi(\theta) \exp\{-\omega T \cdot \mathcal{D}(\theta; \mathbb{P})\} \quad \text{for} \quad \mathcal{D}(\theta; \mathbb{P}) = \mathbb{E}_{X \sim \mathbb{P}}[d(\theta, X)]. \quad (6)$$

Here, the discrepancy-based loss $\mathcal{D}(\theta; \mathbb{P}) = \mathbb{E}_{X \sim \mathbb{P}}[d(\theta, X)]$ allows us to recover theoretical posteriors based on averaging some kind of discrepancy $d : \Theta \times \mathcal{X} \rightarrow \mathbb{R}$ for any measure $\mathbb{P}$. This makes the results more general and more natural to derive. Note that all derived results apply to the $\mathcal{D}_m$ -posterior computed from data points $x_{1:T}$, as we can recover it by considering $d = d_m$, and the corresponding empirical measure $\mathbb{P}_T = \frac{1}{T} \sum_{t=1}^T \delta_{x_t}$. In particular, for $d = d_m$ we have that $\mathcal{D}(\theta; \mathbb{P}_T) = \widehat{\mathcal{D}}_m(\theta)$ so that $\pi_\omega^\mathcal{D}(\theta|\mathbb{P}_T) = \pi_\omega^{\mathcal{D}_m}(\theta|x_{1:T})$ as defined in (4).

The original work of Matsubara et al. (2022b) constructed a proof of robustness for posteriors that did not depend on an averaged loss $\mathcal{D}(\theta; \mathbb{P}) = \mathbb{E}_{X \sim \mathbb{P}}[d(\theta, X)]$, and so the conditions they derive do not exploit this averaged form. Instead, they showed in Lemma 5 of their paper that global bias-robustness holds if

$$\sup_{\theta \in \Theta} \sup_{y \in \mathcal{X}} \left| \frac{d}{d\varepsilon} \mathcal{D}(\theta; \mathbb{P}_{\varepsilon, y})|_{\varepsilon=0}(y, \theta, \mathbb{P}) \right| \pi(\theta) < \infty, \quad \text{and} \quad (7)$$

$$\int_{\Theta} \sup_{y \in \mathcal{X}} \left| \frac{d}{d\varepsilon} \mathcal{D}(\theta; \mathbb{P}_{\varepsilon, y})|_{\varepsilon=0}(y, \theta, \mathbb{P}) \right| \pi(\theta) d\theta < \infty. \quad (8)$$

Clearly, we can simplify this further because our loss is an average. As long as the function d over which the loss is averaged is sufficiently regular, the below result shows that we obtain global bias-robustness.

Proposition B.1. *For each $\theta \in \Theta$. Suppose that π is upper bounded over Θ . If there exists a function $\gamma : \Theta \rightarrow \mathbb{R}$ such that:*

1. $\sup_{y \in \mathcal{X}} |d(\theta, y)| \leq \gamma(\theta)$,
2. $\sup_{\theta \in \Theta} \gamma(\theta) \pi(\theta) < \infty$, and
3. $\int_{\Theta} \gamma(\theta) \pi(\theta) d\theta < \infty$.

Then the posterior influence function PIF($y, \theta, \mathbb{P}$) of $\pi_\omega^\mathcal{D}(\theta|\mathbb{P})$ defined in (6) is bounded over both $\theta \in \Theta$ and $y \in \mathcal{Y}$, so that the $\pi_\omega^\mathcal{D}(\theta|\mathbb{P})$ is globally robust.

Proof. As outlined above, we simply have to show that the above conditions suffice to guarantee (7) and (8). Rewriting the loss function related to the contamination model would be:

$$\mathcal{D}(\theta; \mathbb{P}_{\varepsilon, y}) = \mathbb{E}_{x \sim \mathbb{P}_{\varepsilon, y}}[d(\theta, x)] = (1 - \varepsilon) \mathbb{E}_{x \sim \mathbb{P}}[d(\theta, x)] + \varepsilon \mathbb{E}_{x \sim \delta_y}[d(\theta, x)].$$

Then, differentiating the last expression w.r.t. ε , and evaluating $\varepsilon = 0$, we obtain:

$$\frac{d}{d\varepsilon} \mathcal{D}(\theta; \mathbb{P}_{\varepsilon, y})|_{\varepsilon=0} = \mathbb{E}_{x \sim \mathbb{P}}[d(\theta, x)] + \mathbb{E}_{x \sim \delta_y}[d(\theta, x)]$$

Using Jensen's inequality, we bound the expression $|\frac{d}{d\varepsilon} \mathcal{D}(\theta; \mathbb{P}_{\varepsilon, y})|_{\varepsilon=0}|$ as follows:

$$\begin{aligned} \left| \frac{d}{d\varepsilon} \mathcal{D}(\theta; \mathbb{P}_{\varepsilon, y})|_{\varepsilon=0} \right| &\leq |\mathbb{E}_{x \sim \delta_y}[d(\theta, x)]| + |\mathbb{E}_{x \sim \mathbb{P}}[d(\theta, x)]| \\ &\leq \mathbb{E}_{x \sim \delta_y}[|d(\theta, x)|] + \mathbb{E}_{x \sim \mathbb{P}}[|d(\theta, x)|] \\ &= |d(\theta, y)| + \mathbb{E}_{x \sim \mathbb{P}}[|d(\theta, x)|] \\ &\leq |d(\theta, y)| + \mathbb{E}_{x \sim \mathbb{P}}[\sup_{y \in \mathcal{X}} |d(\theta, y)|] \\ &= |d(\theta, y)| + \sup_{y \in \mathcal{X}} |d(\theta, y)|, \end{aligned}$$